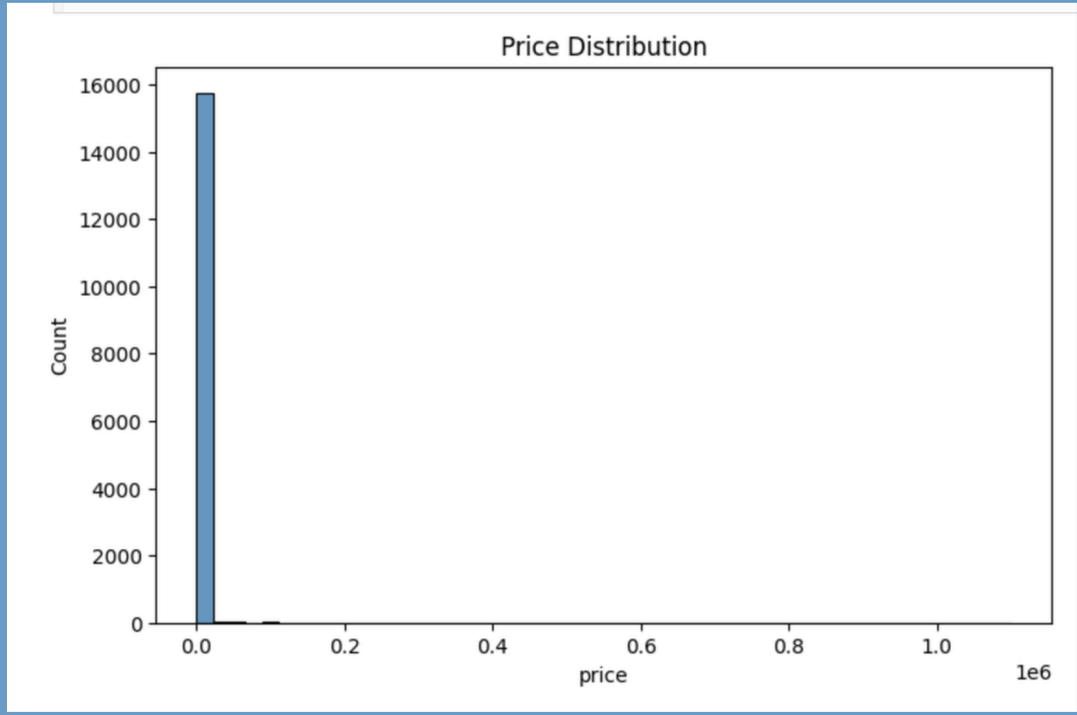
OUTPUTS LIDES



The price distribution is heavily right-skewed, meaning most Airbnb listings fall in the low-price range, while a small number of properties have extremely high prices.

Key Observations

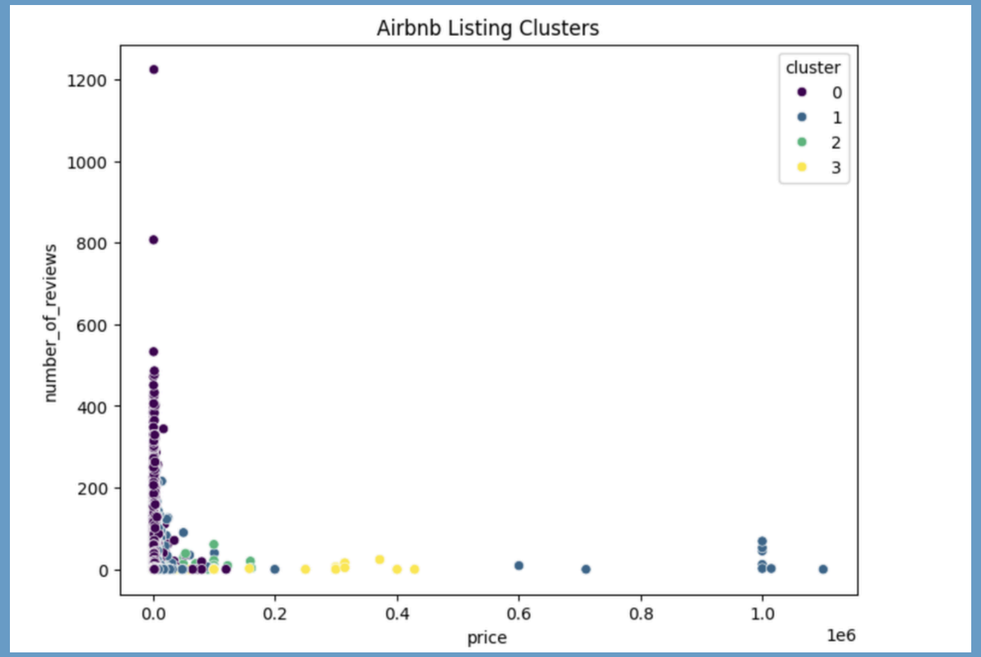
Majority of listings are priced below ₹5000 equivalent

Few listings exceed ₹100,000, creating outliers

This indicates the market is dominated by affordable accommodations

Business Insight

Affordable listings attract the majority of users, while luxury listings represent a niche market segment.



Insight

K-Means clustering divides listings into four distinct market segments based on price and engagement.

Cluster Interpretation

Cluster 0

Moderate price listings with high review counts → Popular properties

Cluster 1

Luxury properties with high price and larger capacity

Cluster 2

Listings with high booking success and balanced pricing

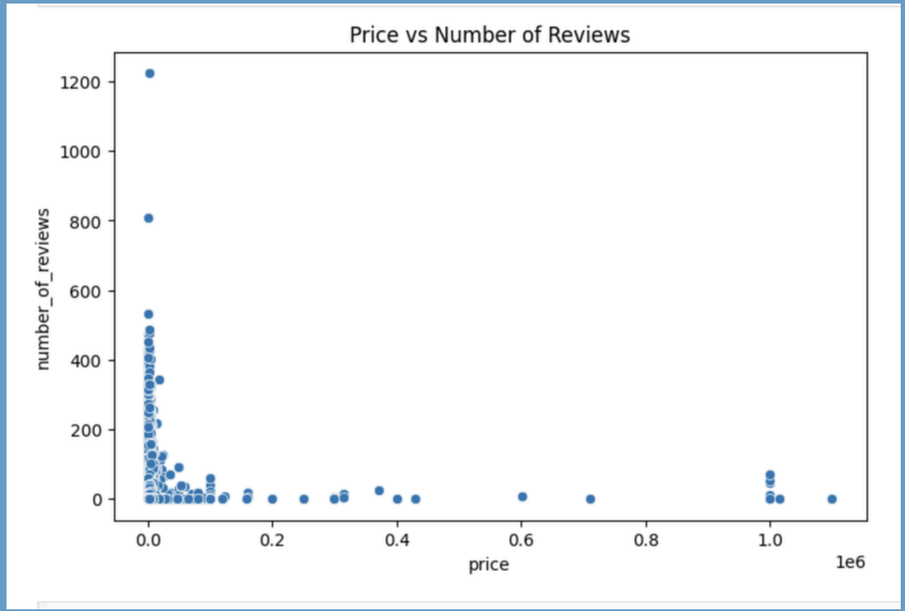
Cluster 3

Listings with low engagement and high availability

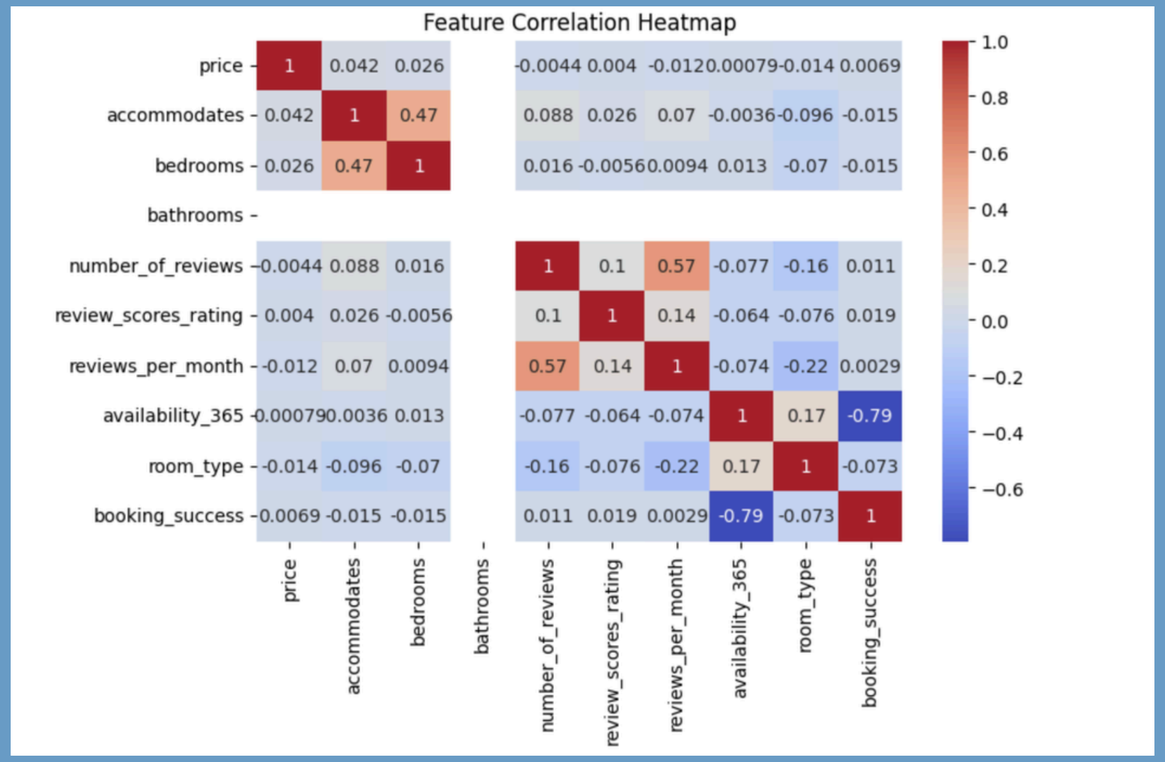
Business Insight

Airbnb listings naturally segment into budget, premium, high-demand, and low-engagement markets.

OUTPUTS LIDES



Insight
There is a weak inverse relationship between price and number of reviews.
Key Observations
Lower-priced listings receive significantly more reviews
Expensive listings tend to have fewer bookings and fewer reviews
Some extremely expensive listings appear as outliers
Business Insight
Competitive pricing increases booking frequency and customer engagement.

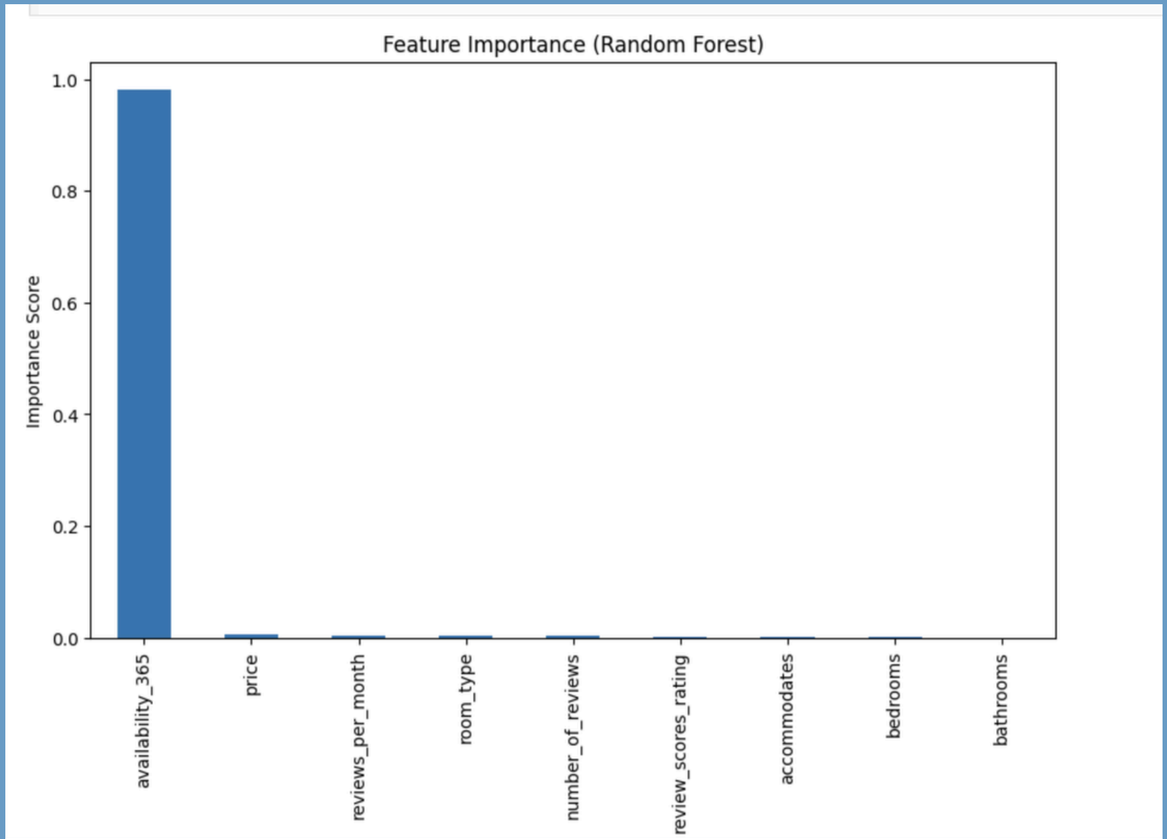


Insight
There is a weak inverse relationship between price and number of reviews.
Key Observations
Lower-priced listings receive significantly more reviews
Expensive listings tend to have fewer bookings and fewer reviews
Insight
The correlation matrix highlights relationships between listing characteristics and booking behavior.
Key Observations

- Accommodates and bedrooms show moderate correlation (≈0.47)
- Number of reviews and reviews per month are strongly related (≈0.57)
- Availability_365 has strong negative correlation with booking success (-0.79)

Business Insight
Properties with lower availability tend to be booked more frequently, indicating higher demand.
Business Insight
Competitive pricing increases booking frequency and customer engagement.

OUTPUTSLIDES



Insight

The Random Forest model identifies the most influential variables affecting booking success.

Key Findings

Availability_365 dominates the prediction

Other variables like price, reviews, and room type have minor influence

Interpretation

Listings that remain available for fewer days tend to be booked more often, making availability the strongest indicator of demand.